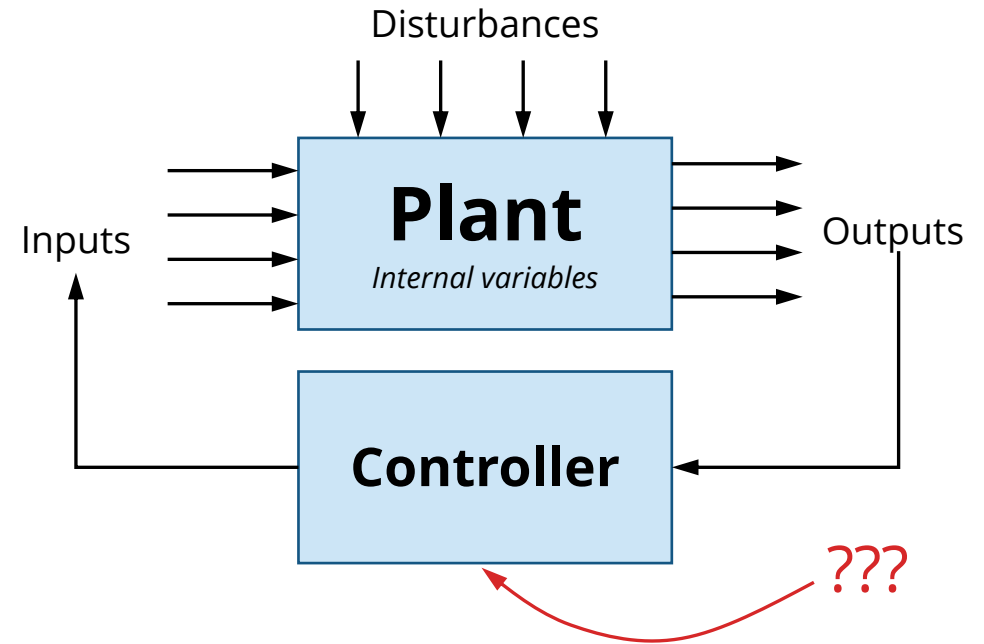




AA 513 Multivariable Control

Lecture 03 // Week 3

Instructor: Prof. Karen Leung

Announcements

- Exercises due Thursday
- TA OH Wed 6PM
- Extra credit for Ed Discussion participation
 - Great discussion so far!
- Released Matlab – Python “Rosetta Stone”
- Ed Discussion
 - Share error message, and python “code snippet”

Last time

- Optimization
 - Gradient descent
 - Convex optimization
 - Frame control problems as (convex) optimization problems
 - Introduced CVXPY

This week

- Lyapunov theory
- Invariant sets
- Control barrier functions
- <https://uw-ctrl.github.io/lmc-book/lectures/control-certificates.html>

Exercises

- Ex4 Equilibria
- Ex5 Lyapunov function
- Ex6 Control Barrier Functions and safety filters

Equilibria

- Given dynamics $\dot{x} = f(x)$, we want to find *equilibrium points*
 - x^{eq} such that $\dot{x} = 0$
- Often define operating points for control design
- Reveal qualitative behavior of nonlinear systems
 - Does the system converge to or diverge from equilibrium points?
- We find equilibrium points because they anchor our understanding of what the system does in steady state and form the basis for analyzing and controlling its behavior.

Stability

- Stability is an important property to characterize.
- Can tell us whether the system's behavior is predictable, safe, and controllable in the face of small perturbations or disturbances.
- If “do nothing”, will the system “stay” or “explode/fall”?
 - E.g., Chemical processes, position of aircraft

Types of stability: Regarding a system I/O

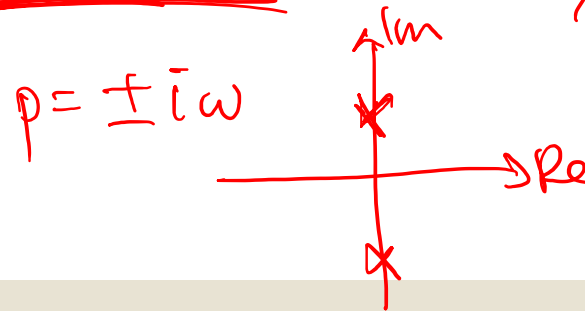
Bounded input bounded output

BIBO stability: A system is BIBO stable if every bounded input produces a bounded output.

- If you feed it a finite signal, it produces a finite response — no unbounded growth or oscillation.*

- BIBO stability check:** For a rational transfer function (LTI systems): ALL poles have negative real parts *all poles in LHP.*

- $G(s) = \frac{1}{s^2 + \omega^2}$ BIBO stable? *poles?*



Types of stability: Regarding equilibria

- **Lyapunov stable**: If trajectory starts close to equilibrium point, then trajectories stay close
 - No guarantees on convergence
- **Asymptotic stability**: If trajectory starts close to equilibrium point, then trajectories eventually converge to the equilibrium point
- **Exponential stability**: If trajectory starts close to equilibrium point, then trajectories converge to the equilibrium point at an exponential rate.

Types of stability: Local vs Global

- **Local stability:** Stability property only holds *near* equilibrium point.
 - Common for nonlinear systems
- **Global stability:** Stability property holds *for all* initial conditions
 - Often harder to prove
 - Easy to prove for linear systems

Contraction theory



Stability of linear systems

- Consider $\dot{x} = Ax$, where solution is: $x(t) = e^{At}x(0)$ *is this stable?*
- Stability analysis based on properties of A
- $A = PDP^{-1} \Rightarrow e^A = Pe^{Dt}P^{-1} = P \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_n t} \end{bmatrix} P^{-1}$
- Asymptotic stability depends on eigenvalues. All real parts *are* negative implies all trajectories will converge to 0 for any initial condition.
- Discrete-time LTI: *λt* Eigenvalues lie inside *$\text{Re}(\lambda) < 0$* of the unit circle *if $\lambda = a + ib$, e^{at} $\rightarrow 0$*

What about nonlinear systems?

- What about $\dot{x} = f(x)$?
- Linear algebra can't save us ☹️
- All nonlinearities are unique

Lyapunov analysis (Rough idea)

- Consider an energy-like function ~~associated with the state trajectory~~ centered about equilibrium

- Energy = 0 implies at ^{stable} equilibrium point

converge
→

- If the energy is **always decreasing** along system trajectories, then the trajectories must approach the equilibrium

asym/exp
stability

Lyapunov
stable
→

- If the energy is **non-increasing** along system trajectories, then the trajectories do not diverge from equilibrium

- Works for nonlinear systems!

$$\dot{x} = -x \Rightarrow x(t) = Ae^{-t}$$

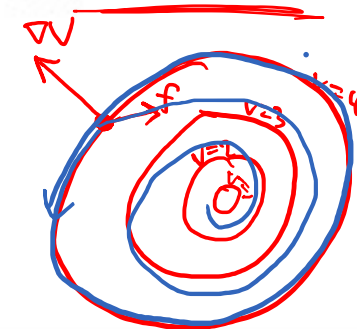
$$\dot{x} = -\alpha x \Rightarrow x(t) = Ae^{-\alpha t}$$

Lyapunov theorem

i Theorem (Lyapunov function)

Given a general nonlinear dynamical system $\dot{x} = f(x)$, $x \in \mathcal{X} \subset \mathbb{R}^n$. Without loss of generality, let $x = 0$ be an equilibrium state. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar function that is continuous, and has continuous first derivatives. Then the origin $x = 0$ is Lyapunov stable if:

- $V(0) = 0$ "resting point"
- $V(x) > 0 \forall x \in \mathcal{X} \setminus \{0\}$
- $\dot{V}(x) = \nabla V(x)^T f(x) \leq 0 \forall x \in \mathcal{X}$ lse where $\dot{V}(x) \leq 0$



How does last equation change for Lyapunov/asymptotic/exponential stability?

Finding a Lyapunov function is not always easy.

$$\dot{V} \leq 0 \quad \dot{V} < 0 \quad \dot{V} \leq -\alpha V$$

Lyapunov stability theorem of linear systems

- For a linear system $\dot{x} = Ax$, consider a *quadratic* Lyapunov candidate

$$V(x) = x^T P x, \quad P \succ 0$$

P positive definite and symmetric

$$\hookrightarrow V(0)=0, V(x)>0 \quad x \in \mathbb{R}^n \setminus \{0\}$$

- Compute $\dot{V}(x(t))$

$$(Ax)^T = x^T A^T \quad \underline{\quad} = \dot{x}^T P x + x^T P \dot{x}$$

$$= x^T \underline{A^T P} x + x^T \underline{P A} x$$

Choose a Q, and solve for P

$$\underline{x^T (A^T P + P A) x < 0} \Rightarrow \underline{A^T P + P A + Q = 0}, Q \succ 0$$

asymptotic stability

Discrete time Lyapunov equation

[exercise] What is the discrete-time Lyapunov equation?

$$\underline{V(x_t) = x_t^T P x_t}, \quad \underline{x_{t+1} = A x_t} \quad \underline{V(x_{t+1}) - V(x_t) < 0} \quad V(x_{t+1}) < V(x_t)$$

$$\underline{x_{t+1}^T P x_{t+1}} - x_t^T P x_t < 0$$

$$\underline{x_t^T A^T P A x_t} - \underline{x_t^T P x_t} < 0$$

$$x_t^T (A^T P A - P) x_t < 0$$

$$\Rightarrow \boxed{A^T P A - P + Q = 0} \text{ for } Q > 0$$

Linear dyn.
Discrete-time.

What if there is a control input?

- What happens when $\dot{x} = f(x, \underline{u})$? The control input could make the system stable/unstable.
- **Pole placement.** For linear systems, $\dot{x} = Ax + Bu$, consider full state feedback linear controller $\underline{u} = -Kx$
 - Closed loop system: $\dot{x} = (A - BK)x$ $A_{cl} = A - BK$
 - Choose K that makes the closed-loop eigenvalues have negative real-parts
 - (See exercise)

$$K \in \mathbb{R}^{m \times n} \quad K = \begin{bmatrix} K_{11} & K_{12} & \dots & K_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ K_{m1} & K_{m2} & \dots & K_{mn} \end{bmatrix}$$

What if there is a control input?

• Control Lyapunov Function

- Given x , choose $u \in U$ such that

$$\nabla V(x)^T f(x, u) \leq -\alpha V(x) \quad (\text{or } 0, \text{ or } <)$$

$$0 < 0 \leq 0$$

➔ What happens if dynamics are control affine?

- Ensure feasible control for all states. May need to adjust α

$$U_{\text{stable}} = \{u \in U \mid \nabla V(x)^T f(x, u) \leq -\alpha V(x)\}$$

$$f(x, u) = \underbrace{f_0(x)} + \underbrace{B(x)u} \Rightarrow \nabla V(x)^T (f_0(x) + B(x)u) \leq -\alpha V(x)$$

$$\nabla V(x)^T f_0(x) + \nabla V(x)^T B(x)u \leq -\alpha V(x)$$

- Intuition:** Pick control satisfying the constraint $\dot{V}(x) \leq -\alpha(V(x))$

$$C_1 u \leq C_0$$

Control Lyapunov Function example

• $\dot{x}_1 = x_2, \dot{x}_2 = -x_1 + u, V(x) = \frac{1}{2}(x_1^2 + x_2^2), \dot{V}(x) \leq 0$ *asymptotically stable.*

$$\dot{V}(x) = \nabla V(x)^T f(x, u) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ -x_1 + u \end{bmatrix} = \cancel{x_1 x_2} - \cancel{x_1 x_2} + u x_2$$

$$\nabla V(x) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \underline{u x_2} < 0$$

pick $u = -\underline{k x_2}, k > 0$

$$= \underline{-k x_2^2} < 0 \quad \checkmark$$

$$u = \text{clip}(-k x_2, u_{\min}, u_{\max})$$

Exercises

Then control barrier functions + exercises

Control invariant sets

- **Forward invariant set:** A set such that if system starts inside, it will remain inside.

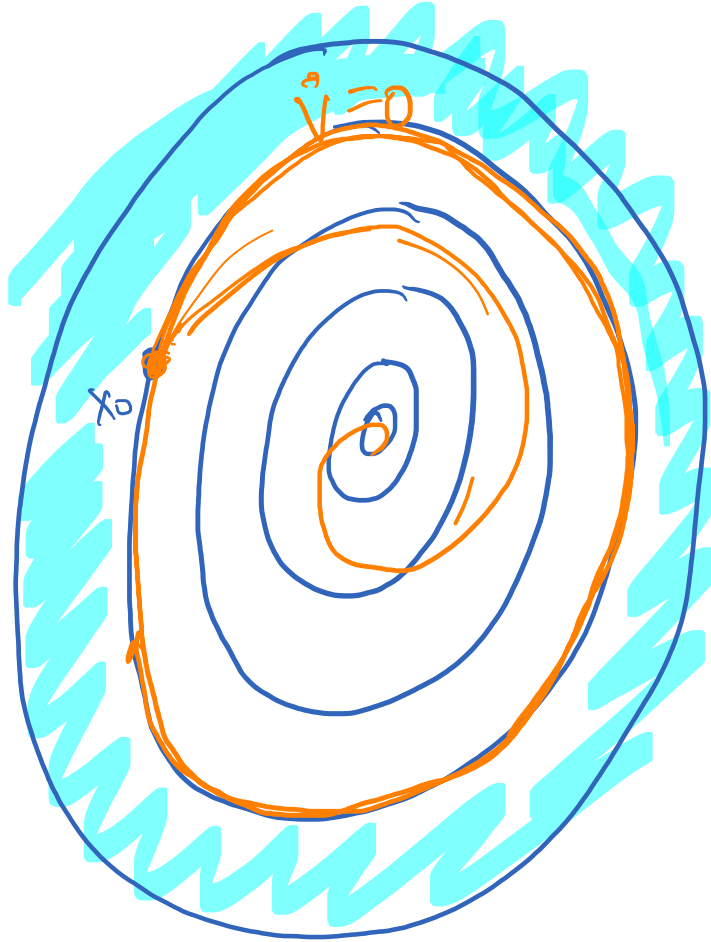
- Pendulum



- **Control invariant set:** Same as above, but “there exists a control” such that...
- Instead of converging to state/point, can we remain inside a set?

Level sets of (control)Lyapunov functions are (control) invariant sets

$$\dot{V}(x) \leq 0$$



$$f(x, u) = f_0(x) + B(x)u \quad \nabla b(x)^T (f_0(x) + Bu) \geq -\alpha(b(x))$$

$\Rightarrow$ linear in u $C_1 u \geq C_0$

Control Barrier Functions

i Definition (Control barrier function)

Given a set $\mathcal{S} = \{x \mid b(x) \geq 0\}$ where $b : \mathbb{R}^n \rightarrow \mathbb{R}$. Then b is a control barrier function for a system $\dot{x} = f(x, u)$ if there exists a class $\mathcal{K}$ function α such that

$$\max_{u \in \mathcal{U}} \nabla b(x)^T f(x, u) \geq -\alpha(b(x)), \quad \forall x \in \mathcal{S} \quad (3)$$

$\dot{b}(x) \geq -\alpha(b(x))$



- Barrier function describes set to keep system inside of.
- Constraint that *bounds the rate* at which the system approaches the boundary.

$\dot{b}(x) \geq 0$ when $b(x) = 0$

- What happens when dynamics are control affine?



$\dot{b}(x) \geq -10$

$\dot{b}(x) \geq -1 \cdot 0 = 0$

$$\|u\|_2^2 = u_1^2 + u_2^2 + \dots + u_m^2$$

CBF safety filter

- We can use CBFs to *filter*, or adjust, the desired control to ensure safety (and stability) conditions are satisfied.
- For current state x , and desired control input $u^{desired}$ (e.g., from high-level planner)

QP

$$\begin{aligned} & \min_u \quad \underbrace{\|u - u^{desired}\|_2^2}_{\text{quadratic}} \\ \text{s.t.} \quad & \underbrace{\nabla b(x)^T (f_0(x) + B(x)u) \geq -\alpha(b(x))}_{\text{linear}} \quad \text{CBF constant} \\ & \underbrace{|u| \leq u^{limit}}_{\text{linear}} \\ & \underbrace{u_{min} \leq u \leq u_{max}}_{\text{linear}} \end{aligned}$$

Question: What type of optimization problem this is? QP

CBF-CLF filter

- We can add a CLF constraint as well!



$\dot{v} \neq 0$

$\dot{b} \geq -\alpha b$

$$\min_u \|u - \underline{u^{desired}}\|_2^2$$

s.t. $\nabla b(x)^T (f_0(x) + B(x)u) \geq -\alpha_{CBF}(b(x))$

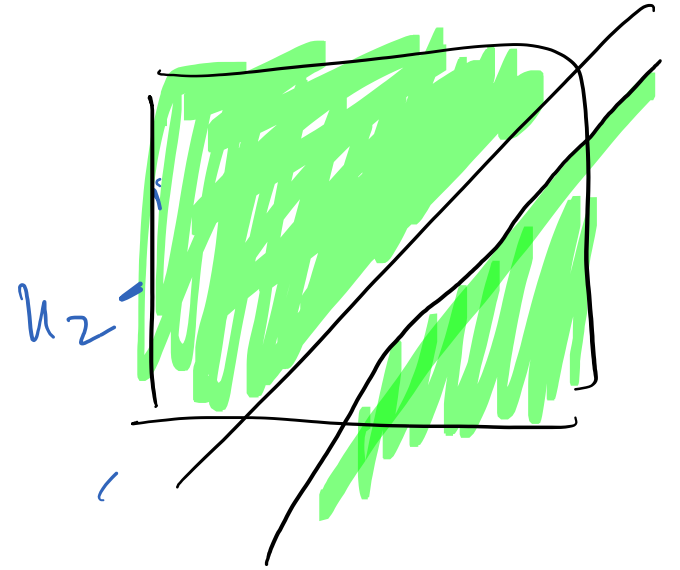
CBF

$\nabla V(x)^T (f_0(x) + B(x)u) \leq -\alpha_{CLF}(b(x))$

CLF

linear in u

$|u| \leq \underline{u^{limit}}$



Problems? the intersection of the CBF & CLF constraint can be empty

Further readings

- See course notes:
 - <https://uw-ctrl.github.io/lmc-book/lectures/control-certificates.html#other-topics-and-extra-readings> ←
- Use neural networks to parameterize functions
- Learn CBFs from demonstrations
- Used to filter neural network policies
- Used as inductive bias in learning-based models
- Professor Aaron Ames from Caltech:
https://www.youtube.com/watch?v=ZC3T_P_8xpE